

RENEWABLE ENERGY AND GREEN HYDROGEN

POWER METRIC FIREWALL → ENERGY CATEGORY MAP → GRID INTEGRATION RAIL → PROJECT STATUS LADDER → ELECTROLYSIS PATHWAY → HYDROGEN CLAIM STACK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00POWER METRIC FIREWALL

POWER METRIC FIREWALLINSTALLED CAPACITYNAMEPLATE POWER CAPABILITYELECTRICAL ENERGY PRODUCED OVER TIMECAPACITY SHAREPORTFOLIO CAPABILITY, NOT OUTPUT SHARERELIABLE SUPPLYTIMING, FLEXIBILITY AND NETWORK MATTER

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
INSTALLED CAPACITY → nameplate power capability	CAPACITY SHARE → portfolio capability, not output share	RULE → never convert megawatts into generation without evidence
GENERATION → electrical energy produced over time	RELIABLE SUPPLY → timing, flexibility and network matter	MUST REMEMBER: Renewable-energy transition must connect resource, installed capacity,...

CORE CLAIM

- INSTALLED CAPACITY → nameplate power capability
- GENERATION → electrical energy produced over time

MECHANISM

- CAPACITY SHARE → portfolio capability, not output share
- RELIABLE SUPPLY → timing, flexibility and network matter

CONSEQUENCE / LIMIT

- RULE → never convert megawatts into generation without evidence
- MUST REMEMBER: Renewable-energy transition must connect resource, installed capacity,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never convert megawatts into generation without evidence.

01

STAGE 01ENERGY CATEGORY MAP

ENERGY CATEGORY MAPSOLAR WIND HYDRO BIOMASS GEOTHERMALRENEWABLE SOURCES PLUS OTHER NON-FOSSIL SOURCESGENERATED ENERGY OUTPUTMANUFACTURED ENERGY CARRIERSOURCE, OUTPUT AND CARRIER ARE DIFFERENT CATEGORIES

RENEWABLE → solar wind hydro biomass geothermal

NON-FOSSIL → renewable sources plus other non-fossil sources

ELECTRICITY → generated energy output

HYDROGEN → manufactured energy carrier

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
RENEWABLE → solar wind hydro biomass geothermal	ELECTRICITY → generated energy output	RULE → source, output and carrier are different categories
NON-FOSSIL → renewable sources plus other non-fossil sources	HYDROGEN → manufactured energy carrier	RENEWABLE → solar wind hydro biomass geothermal

CORE CLAIM

- RENEWABLE → solar wind hydro biomass geothermal
- NON-FOSSIL → renewable sources plus other non-fossil sources

MECHANISM

- ELECTRICITY → generated energy output
- HYDROGEN → manufactured energy carrier

CONSEQUENCE / LIMIT

- RULE → source, output and carrier are different categories
- RENEWABLE → solar wind hydro biomass geothermal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → source, output and carrier are different categories.

02

STAGE 02GRID INTEGRATION RAIL

GRID INTEGRATION RAILVARIABLE GENERATIONFORECAST RESOURCE-DEPENDENT OUTPUTMOVE POWER FROM RESOURCE TO DEMANDFLEXIBLE DEMAND AND BALANCINGMATCH SYSTEM CONDITIONSSHIFT ENERGY AND PROVIDE SELECTED SERVICESRELIABILITY IS A SYSTEM OUTCOME

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
VARIABLE GENERATION → forecast resource-dependent output	FLEXIBLE DEMAND AND BALANCING → match system conditions	DELIVERY → reliability is a system outcome
TRANSMISSION → move power from resource to demand	STORAGE → shift energy and provide selected services	VARIABLE GENERATION → forecast resource-dependent output

CORE CLAIM

- VARIABLE GENERATION → forecast resource-dependent output
- TRANSMISSION → move power from resource to demand

MECHANISM

- FLEXIBLE DEMAND AND BALANCING → match system conditions
- STORAGE → shift energy and provide selected services

CONSEQUENCE / LIMIT

- DELIVERY → reliability is a system outcome
- VARIABLE GENERATION → forecast resource-dependent output

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE → shift energy and provide selected services.

03

STAGE 03PROJECT STATUS LADDER

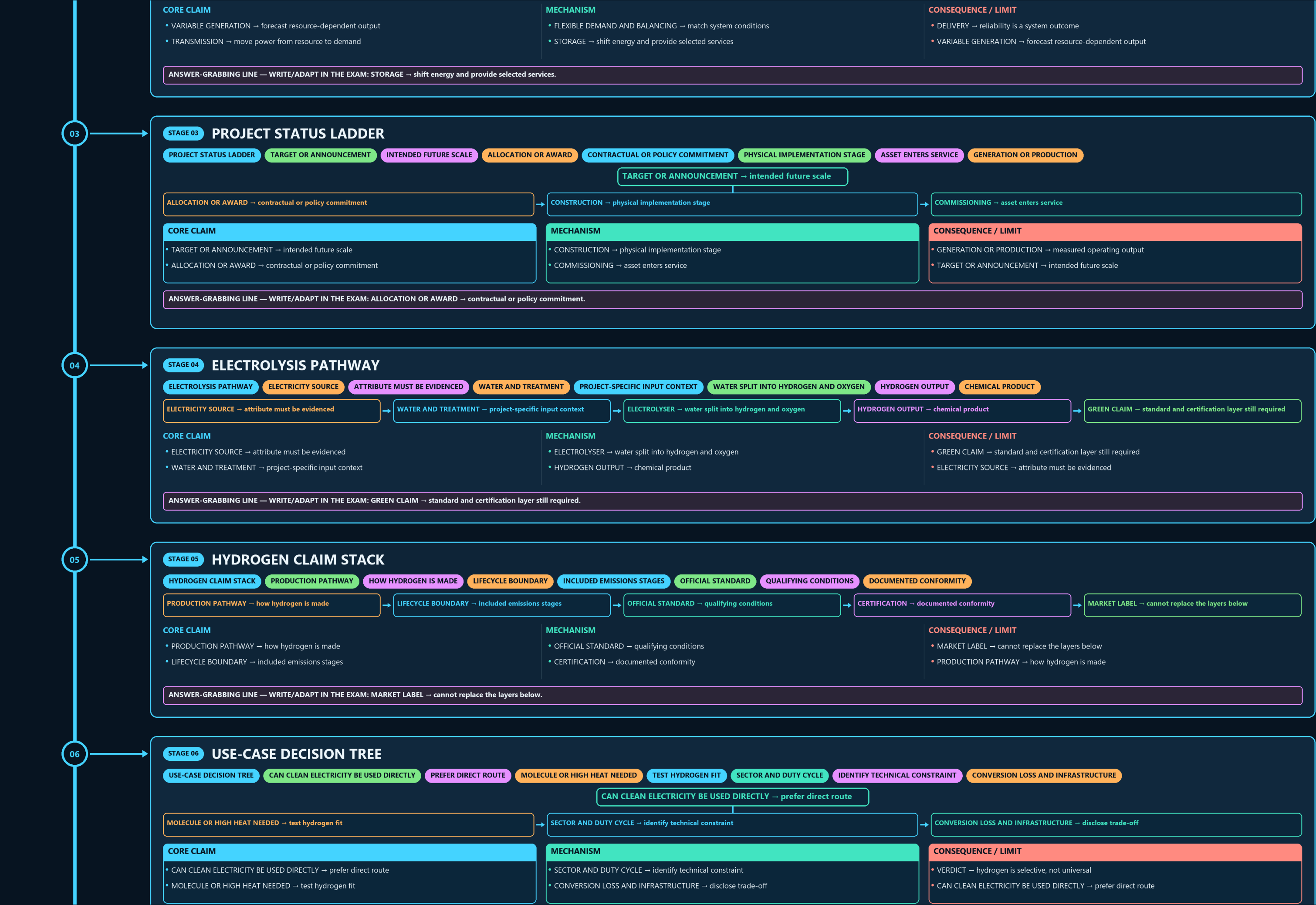
PROJECT STATUS LADDERTARGET OR ANNOUNCEMENTINTENDED FUTURE SCALEALLOCATION OR AWARDCONTRACTUAL OR POLICY COMMITMENTPHYSICAL IMPLEMENTATION STAGEASSET ENTERS SERVICEGENERATION OR PRODUCTION

TARGET OR ANNOUNCEMENT → intended future scale

ALLOCATION OR AWARD → contractual or policy commitment

CONSTRUCTION → physical implementation stage

COMMISSIONING → asset enters service



06

STAGE 06 USE-CASE DECISION TREE

USE-CASE DECISION TREE CAN CLEAN ELECTRICITY BE USED DIRECTLY PREFER DIRECT ROUTE MOLECULE OR HIGH HEAT NEEDED TEST HYDROGEN FIT SECTOR AND DUTY CYCLE IDENTIFY TECHNICAL CONSTRAINT CONVERSION LOSS AND INFRASTRUCTURE

CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit

SECTOR AND DUTY CYCLE → identify technical constraint

CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

CORE CLAIM

- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route
- MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit

MECHANISM

- SECTOR AND DUTY CYCLE → identify technical constraint
- CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

CONSEQUENCE / LIMIT

- VERDICT → hydrogen is selective, not universal
- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR AND DUTY CYCLE → identify technical constraint.

07

STAGE 07 DERIVATIVE CONVERSION MAP

DERIVATIVE CONVERSION MAP BASE ENERGY CARRIER AND FEEDSTOCK NITROGEN-BASED DERIVATIVE METHANOL OR SYNTHETIC FUEL CARBON-CONTAINING DERIVATIVE CONVERSION AND DELIVERY EXTRA ENERGY AND INFRASTRUCTURE PRODUCT TARGETS AND OUTPUTS REMAIN SEPARATE

HYDROGEN → base energy carrier and feedstock

AMMONIA → nitrogen-based derivative

METHANOL OR SYNTHETIC FUEL → carbon-containing derivative

CONVERSION AND DELIVERY → extra energy and infrastructure

CORE CLAIM

- HYDROGEN → base energy carrier and feedstock
- AMMONIA → nitrogen-based derivative

MECHANISM

- METHANOL OR SYNTHETIC FUEL → carbon-containing derivative
- CONVERSION AND DELIVERY → extra energy and infrastructure

CONSEQUENCE / LIMIT

- RULE → product targets and outputs remain separate
- HYDROGEN → base energy carrier and feedstock

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → product targets and outputs remain separate.

08

STAGE 08 WATER AND SITE SCREEN

WATER AND SITE SCREEN FRESHWATER TREATED WATER OR OTHER VERIFIED SUPPLY PRETREATMENT REQUIREMENT BASIN AND LOCAL SCARCITY CONTEXT COMPETING DEMAND ECOLOGICAL AND SOCIAL CLAIMS PROJECT EVIDENCE, NOT A UNIVERSAL WATER NUMBER

SOURCE → freshwater treated water or other verified supply

QUALITY → pretreatment requirement

LOCATION → basin and local scarcity context

COMPETING DEMAND → ecological and social claims

CORE CLAIM

- SOURCE → freshwater treated water or other verified supply
- QUALITY → pretreatment requirement

MECHANISM

- LOCATION → basin and local scarcity context
- COMPETING DEMAND → ecological and social claims

CONSEQUENCE / LIMIT

- VERDICT → project evidence, not a universal water number
- SOURCE → freshwater treated water or other verified supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → project evidence, not a universal water number.

09

STAGE 09 HYDROGEN LOGISTICS CHAIN

HYDROGEN LOGISTICS CHAIN HYDROGEN LEAVES ELECTROLYSER COMPRESSION LIQUEFACTION OR CONVERSION STORAGE AND TRANSPORT VESSEL PIPELINE TANKER OR PORT END USE EQUIPMENT AND SAFETY COMPATIBILITY DELIVERED SERVICE DIFFERS FROM PRODUCTION CAPACITY

PRODUCTION → hydrogen leaves electrolyser

CONDITIONING → compression liquefaction or conversion

STORAGE AND TRANSPORT → vessel pipeline tanker or port

END USE → equipment and safety compatibility

OUTCOME → delivered service differs from production capacity

CORE CLAIM

- PRODUCTION → hydrogen leaves electrolyser
- CONDITIONING → compression liquefaction or conversion

MECHANISM

- STORAGE AND TRANSPORT → vessel pipeline tanker or port
- END USE → equipment and safety compatibility

CONSEQUENCE / LIMIT

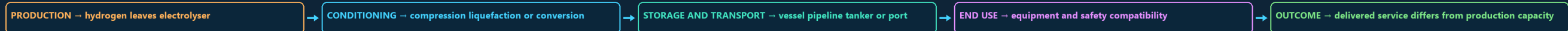
- OUTCOME → delivered service differs from production capacity
- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,...

09

STAGE 09 HYDROGEN LOGISTICS CHAIN

HYDROGEN LOGISTICS CHAIN HYDROGEN LEAVES ELECTROLYSER COMPRESSION LIQUEFACTION OR CONVERSION STORAGE AND TRANSPORT VESSEL PIPELINE TANKER OR PORT END USE EQUIPMENT AND SAFETY COMPATIBILITY DELIVERED SERVICE DIFFERS FROM PRODUCTION CAPACITY



CORE CLAIM

- PRODUCTION → hydrogen leaves electrolyser
- CONDITIONING → compression liquefaction or conversion

MECHANISM

- STORAGE AND TRANSPORT → vessel pipeline tanker or port
- END USE → equipment and safety compatibility

CONSEQUENCE / LIMIT

- OUTCOME → delivered service differs from production capacity
- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,.

10

STAGE 10 MISSION AND SIGHT MAP

MISSION AND SIGHT MAP MISSION TARGET INTENDED FUTURE PRODUCTION SCALE SIGHT MANUFACTURING ELECTROLYSER CAPABILITY SUPPORT SIGHT PRODUCTION HYDROGEN OUTPUT SUPPORT COMMISSIONED OUTPUT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MISSION TARGET → intended future production scale	SIGHT PRODUCTION → hydrogen output support	CLIMATE OUTCOME → separate lifecycle result
SIGHT MANUFACTURING → electrolyser capability support	COMMISSIONED OUTPUT → later measured evidence	EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

CORE CLAIM

- MISSION TARGET → intended future production scale
- SIGHT MANUFACTURING → electrolyser capability support

MECHANISM

- SIGHT PRODUCTION → hydrogen output support
- COMMISSIONED OUTPUT → later measured evidence

CONSEQUENCE / LIMIT

- CLIMATE OUTCOME → separate lifecycle result
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

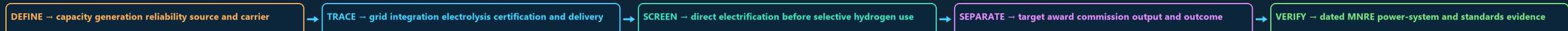
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,.

11

STAGE 11 ENERGY-TRANSITION ANSWER SPINE

ENERGY-TRANSITION ANSWER SPINE CAPACITY GENERATION RELIABILITY SOURCE AND CARRIER GRID INTEGRATION ELECTROLYSIS CERTIFICATION AND DELIVERY DIRECT ELECTRIFICATION BEFORE SELECTIVE HYDROGEN USE TARGET AWARD COMMISSION OUTPUT AND OUTCOME

DATED MNRE POWER-SYSTEM AND STANDARDS EVIDENCE



CORE CLAIM

- DEFINE → capacity generation reliability source and carrier
- TRACE → grid integration electrolysis certification and delivery

MECHANISM

- SCREEN → direct electrification before selective hydrogen use
- SEPARATE → target award commission output and outcome

CONSEQUENCE / LIMIT

- VERIFY → dated MNRE power-system and standards evidence
- DEFINE → capacity generation reliability source and carrier

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated MNRE power-system and standards evidence.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT LARGE-SCALE SOLAR AND WIND INFRASTRUCTURE REQUIRES SUBSTANTIAL LAND AREA BALANCED FRAMING THIS IS NOT A CASE WHERE EITHER RENEWABLE ENERGY BATTERY-STORAGE COSTS HAVE BEEN FALLING BUT REMAIN A SUBSTANTIAL ANALYTICAL POINT

INDIA'S RENEWABLE-CAPACITY-ADDITION TARGETS (500 GW NON-FOSSIL BY GREEN-HYDROGEN PRODUCTION-COST FIGURES AND ELECTROLYSER-CAPACITY FIGURES CHANGE RAPIDLY

OPTIONAL SOURCE DEPTH

- CAUTION: Large-scale solar and wind infrastructure requires substantial land area and, for wind
- CAUTION: Balanced framing: this is not a case where either renewable energy or species
- CAUTION: Battery-storage costs have been falling but remain a substantial cost component of a
- CAUTION: Analytical point: India's renewable-capacity-addition targets (500 GW non-fossil by

ADVANCED DEBATE

- CAUTION: Green-hydrogen production-cost figures and electrolyser-capacity figures change rapidly
- CAUTION: Progress toward the National Green Hydrogen Mission's 5 MMT/year 2030 target and the
- CAUTION: Quantifying the precise trade-off between renewable-infrastructure expansion and
- Solar and wind generation are variable/non-dispatchable, requiring storage and grid

LEGACY / NUANCE

- Green hydrogen's cost-competitiveness depends on both falling electrolyser/renewable
- The Supreme Court has engaged with the tension between renewable-energy transmission
- The SIGHT Programme's dual incentive structure (electrolyser manufacturing plus
- NOT: India's renewable-energy challenge today is primarily about building enough generation

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.